

Chanyoung Moon

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RESEARCH INTERESTS

Computer Architecture, Hardware–Software Co-design, and Domain-Specific Accelerators

EDUCATION

- **University of Wisconsin–Madison** *Sep. 2025 – May 2026*
Visiting Exchange Student, Electrical and Computer Engineering
– GPA: 3.94/4.0
Madison, WI, USA
- **Korea University** *Mar. 2021 – Aug. 2027 (Expected)*
B.E. in Electrical Engineering
– Overall GPA: 4.44/4.5 (Upper-Division: 4.50/4.5), Major GPA: 4.46/4.5
– Leave of absence for mandatory military service: Apr. 2023 – Oct. 2024
Seoul, South Korea

PUBLICATIONS

- [C.1] Chanyoung Moon, Travis Pence, Daisuke Yamada, Jiaqi Mo, Yanxin Zhang, Vikas Singh, Karthikeyan Sankaralingam. **FORGE: Procedural Weight Generation for Sparse MoE Inference**. ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2027).
- [C.2] Travis Pence, Daisuke Yamada, Jiaqi Mo, Chanyoung Moon, Karthikeyan Sankaralingam, Vikas Singh. **Fast Sandwich Products in Clifford Algebra**. Advances in Neural Information Processing Systems (NeurIPS 2026).

RESEARCH EXPERIENCE

- **Vertical Research Group, University of Wisconsin–Madison** *Oct. 2025 – Present*
Undergraduate Researcher (Advisor: Prof. Karthikeyan Sankaralingam)
Madison, WI, USA
 - **Closed-Loop Neural–Machine Interface System Simulation**
Leading simulator development for a simulation-first BCI/exoskeleton co-design project; modeled the sensorimotor loop as a multi-rate dependency graph under a sub-100 ms budget to identify architectural bottlenecks.
 - **3D Gaussian Splatting Rendering Acceleration**
Reproduced GEMM-GS (DAC 2026) as a baseline, profiled it on H100, and identified the sequential alpha-blending stage as the dominant remaining bottleneck; prototyping parallelization of its dependency chain on tensor cores.
 - **Fast Sandwich Products in Clifford Algebra**
Contributed to the GPU kernel for the rotor sandwich product with bank-conflict-free shared-memory access, yielding up to $64\times$ wall-clock and $35\times$ memory improvements over existing implementations ([C.2]).
 - **FORGE: Procedural Weight Generation for Sparse MoE Inference**
 - Led the project and wrote the first-author paper ([C.1]).
 - Designed the Rotor Generation Unit (RGU), an on-SM accelerator that regenerates MoE expert weights on the fly, trading HBM weight traffic for on-chip computation; exposed three axes of data parallelism in the generation algorithm and mapped them onto the datapath.
 - Implemented and verified the RTL in SystemVerilog with cocotb and Verilator, synthesized for area and timing, and built a system-level simulator to project end-to-end speedup.
 - Integrated RGU into Accel-Sim for cycle-level GPU simulation after submission.
 - Cut parameter memory by 28–29% and raised system throughput $1.3\text{--}2.8\times$ at $\sim 0.9\%$ GPU SM area (7 nm).
- **Compiler & Microarchitecture Lab (COMMIT), Korea University** *Jul. 2022 – Sep. 2022*
Undergraduate Research Intern (Advisor: Prof. Seon Wook Kim)
Seoul, South Korea
 - Implemented and verified RISC-V R-Type instruction modules in Verilog HDL on FPGA; published at IEIE Autumn Conference 2022.

HONORS AND AWARDS

- **Minister of Trade, Industry and Resources Award** *Sep. 2026*
1st place among 213 scholars for outstanding academic and research achievement through the Korea–U.S. Advanced Fields Youth Exchange Program
- **NRF Undergraduate Research Fellowship** *Sep. 2026 – Feb. 2027*
National Research Foundation of Korea (for outstanding STEM undergraduates pursuing research careers)
- **3rd Korea–U.S. Advanced Fields Youth Exchange Scholarship (~\$18,000)** *2025 – 2026*
Ministry of Trade, Industry and Resources, Republic of Korea
- **Endowment Scholarship (~\$3,700, Awarded to Top Student)** *May 2025*
School of Electrical Engineering, Korea University
- **Academic Excellence Scholarship (~\$11,200 total)** *Spring 2022, Spring 2025, Fall 2025*
Korea University (awarded for 4.5/4.5 GPA)
- **Guwon Scholarship Foundation Award (~\$1,800)** *2022*
For academic excellence
- **Dean’s Award** *Fall 2021, Fall 2022, Spring 2025*
Korea University (awarded for 4.5/4.5 semester GPA)
- **Semester Highest Honors** *5 semesters, 2021 – 2025*
Korea University

OTHER EXPERIENCE AND ACTIVITIES

- **Defense Communication Command, Republic of Korea Army** *Apr. 2023 – Oct. 2024*
Microwave Communications Operator (Squad Leader) *Daejeon, South Korea*
 - Served as Squad Leader (Jun.–Aug. 2024); participated in UFS (ROK–US Combined Forces Exercise) in Aug. 2023 and Aug. 2024.
- **Young Engineers Honor Society (YEHS), Seoul, South Korea** *May 2025 – Present*
 - Dean-recommended national honor society under the **National Academy of Engineering of Korea (NAEK)** for top engineering students.
- **Woonhwa Club, Korea University** *Apr. 2021 – Aug. 2022*
Vice President, 53rd Cohort *Seoul, South Korea*
 - Taught weekly science classes and coordinated operations of the *Bandi Youth Study Room*, a club-run tutoring program for middle-schoolers from underserved households.

REFERENCES

- **Karthikeyan Sankaralingam**, UW–Madison, Professor; NVIDIA, Principal Research Scientist karu@cs.wisc.edu
- **Jeongseob Ahn**, Korea University, Associate Professor jsahn@csl.korea.ac.kr
- **Seon Wook Kim**, Korea University, Professor seon@korea.ac.kr

SKILLS

- **Programming & Hardware Description:** C, C++, CUDA, Python, Verilog, SystemVerilog
- **EDA & Synthesis Tools:** Xilinx Vivado, ModelSim, Intel Quartus, Yosys, Synopsys Design Compiler
- **Performance Analysis & Simulation:** Accel-Sim, CACTI, NVIDIA Nsight Compute (NCU)
- **HDL Verification:** cocotb, Icarus Verilog, Verilator